

PHYSIOLOGY OF PAIN, PAIN PATHWAYS & OROFACIAL PAIN



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LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- Define pain & classify different types of pain(e.g. acute pain, chronic pain, visceral pain, somatic pain & referred pain).
- Explain the physiological role of pain as a protective mechanism.
- Explain the mechanism of pain transmission to the brain.
- Differentiate between fast & slow pain pathways.
- Describe different types of pain fibers(A-delta fibers & type C fibers) & the neurotransmitters released by these fibers.
- Describe mechanism of pain modulation(ascending & descending pain pathways & gate control theory).
- Define orofacial pain & classify odontogenic & non-odontogenic types.
- Identify common conditions of oro-facial pain(dental caries, pulpitis, TMJ disorders, trigeminal neuralgia, sinus related pain).

INTRODUCTION

- Many ailments of the body cause pain.
- Although pain is unpleasant, it has protective benefits e.g. when the skin becomes painful due to ischemia, the person normally shifts weight subconsciously.
- A person who has lost the pain sense (e.g. spinal cord injury) fails to feel pain & therefore fails to shift which results in total breakdown & desquamation of skin at the sites of pressure.
- The ability of a physician to diagnose different diseases depends to a great extent on his knowledge of different qualities of pain.

What is pain?

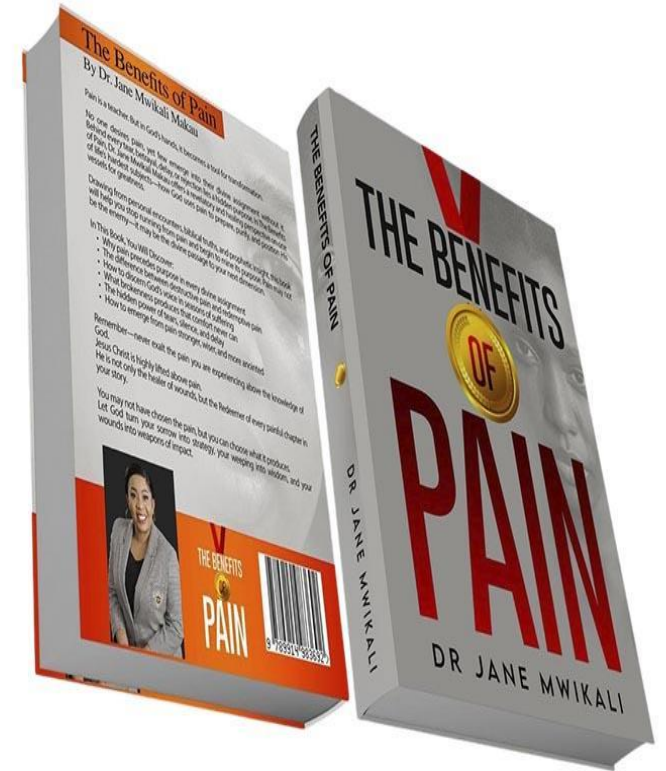
Pain is defined as an unpleasant & emotional experience associated with or without actual tissue damage.

- ❑ Pain is produced by real or potential injury to the body.
- ❑ Pain sensation can be described in many ways like sharp, pricking, electrical, dull, shooting, cutting, stabbing etc.
- ❑ Pain produced by fire can be expressed as burning sensation.
- ❑ Pain produced by severe sustained contraction of skeletal muscles is expressed as cramps.



Although feeling pain is an unpleasant condition, it has some benefits like:

- ❑ Pain gives warning signal about the existence of a problem or threat.
- ❑ It creates awareness of an injury.
- ❑ Pain prevents further damage by causing reflex withdrawal of the body from the source of injury.
- ❑ Pain forces the person to rest or to minimize activities enabling rapid healing of the injured part.
- ❑ Pain urges the individual to take treatment & prevent further major damage.



CLASSIFICATION OF PAIN

FAST PAIN

- Fast pain is the first sensation whenever a pain stimulus is applied.
- Fast pain can also be described as sharp pain, pricking pain, acute pain, electric pain.
- Fast pain is felt within about 0.1 second after a pain stimulation.
- This type of pain is felt when a needle pricks into the skin or when the skin is cut with a knife or burn acutely.
- Fast pain is also felt the skin is subjected to electric shock.
- Fast sharp pain is not felt in most deep tissues of the body.

SLOW PAIN

- Fast pain is followed by the slow pain, which is experienced as a dull, diffused unpleasant pain.
- This type of pain can be described as slow burning pain, aching pain, throbbing pain, nauseous pain, chronic pain.
- This type of pain is usually associated with tissue destruction.
- Slow pain can occur both in the skin & in almost any deep tissue or organ.

ACUTE PAIN

- Acute pain is a sudden, sharp & intense(severe) pain of short duration.
- Generally it is localized in a small area before spreading to neighbouring areas.
- Acute pain is usually caused by injury to tissues like muscles, bone or visceral organs.
- Injury stimulates nociceptors(pain receptors).
- Usually it is treated by medications.

CHRONIC PAIN

- Chronic pain is the intermittent or continuous pain with different intensities.
- This type of pain lasts for longer periods.
- Chronic pain is usually associated with a long-term illness such as osteoarthritis.
- Chronic pain is generally resistant to medical treatment.

SOMATIC PAIN

Somatic pain is the pain arising due to the stimulation of nociceptors located on skin, mucus membrane & deeper tissues such as bones, muscles, ligaments, tendons, joints & connective tissue.

VISCERAL PAIN

- Visceral pain is caused by stimulation of pain receptors situated in visceral organs.
- Visceral pain is generally poorly localized.
- Substances such as bradykinin & proteolytic enzymes stimulate the pain receptors of viscera.
- Visceral pain can be produced by chemical substances like acidic gastric juice that leak from ruptured ulcers into the peritoneal cavity.
- Spastic contractions of smooth muscles in the gastrointestinal tract & other hollow organs of viscera can also cause visceral pain by stimulating free nerve endings.
- Over-distention of hollow organs also causes visceral pain.

NEUROPATHIC PAIN

Neuropathic pain occurs due to dysfunction or damage, compression or disease of nervous structures.

Neuropathic pain can be of two types:

1. **Peripheral neuropathic pain:** caused by disease or lesion of peripheral nerve or dorsal root or dorsal root ganglion.
2. **Central neuropathic pain:** occurs due to disease or lesion of structures in brain or spinal cord.

REFERRED PAIN

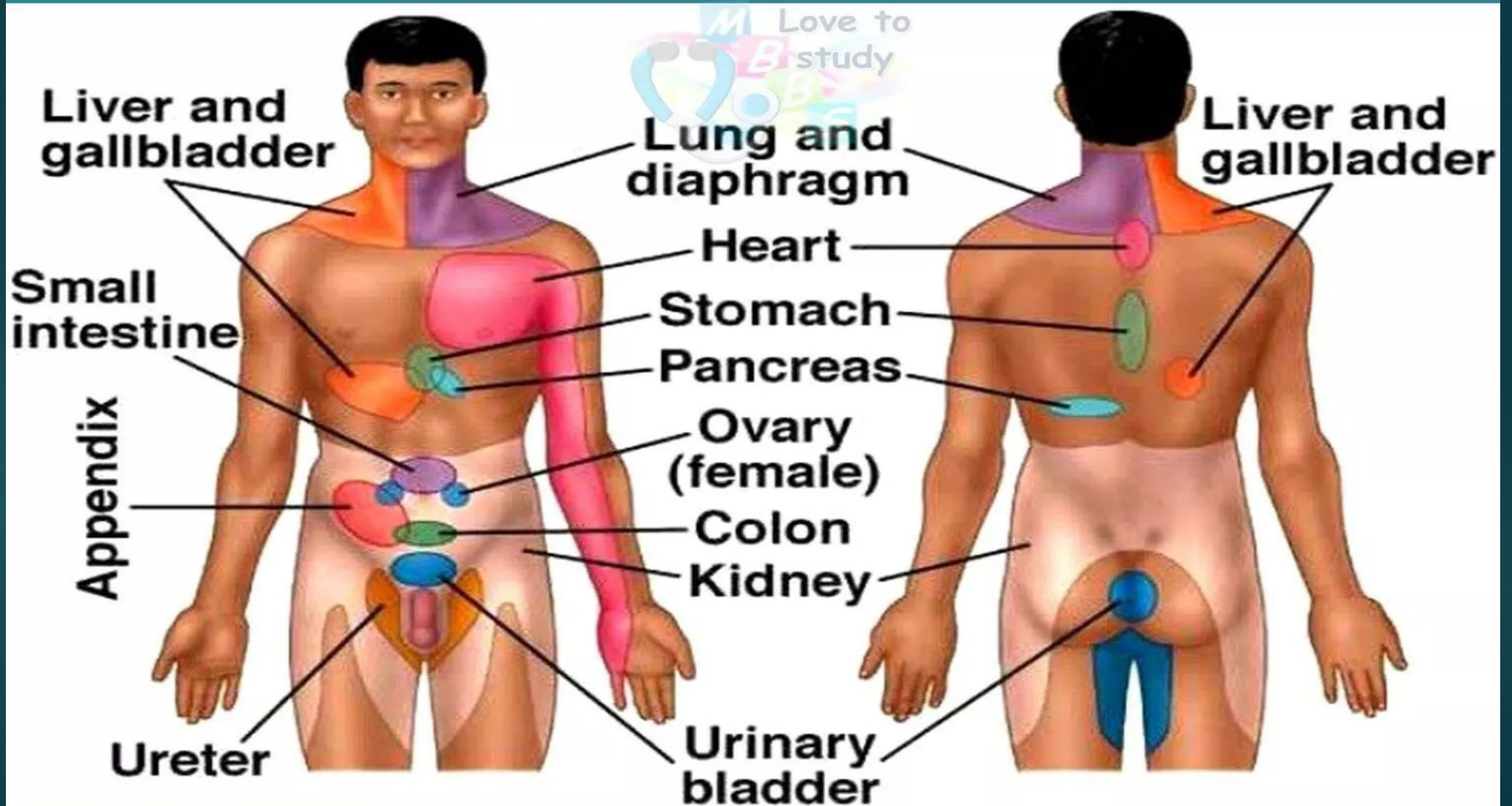
Referred pain can be defined as the pain perceived at a site adjacent to or away from the site of origin.

Knowledge of different types of referred pain is important in clinical diagnosis because in many visceral ailments the only clinical sign is referred pain. Deep pain & some visceral pain are referred to other areas but superficial pain is not referred.

EXAMPLES OF REFERRED PAIN

- ❑ Cardiac pain is felt at the inner part of left arm & left shoulder
- ❑ Pain in ovary is referred to umbilicus
- ❑ Pain from testis is felt in the abdomen
- ❑ Pain in diaphragm is referred to shoulder
- ❑ Renal pain is referred to loin
- ❑ Pain in gallbladder is referred to epigastric region

A Map of Referred Pain



MECHANISM OF REFERRED PAIN (dermatomal rule)

According to dermatomal rule,

pain is referred to a structure which is developed from the same dermatome from which the pain producing structure is developed.

- A dermatome includes all the structures or parts of the body which are innervated by afferent nerve fibres of same dorsal root.
- For example, the heart & inner aspect of left arm originate from the same dermatome so the pain in heart is referred to left arm.

PAIN RECEPTORS

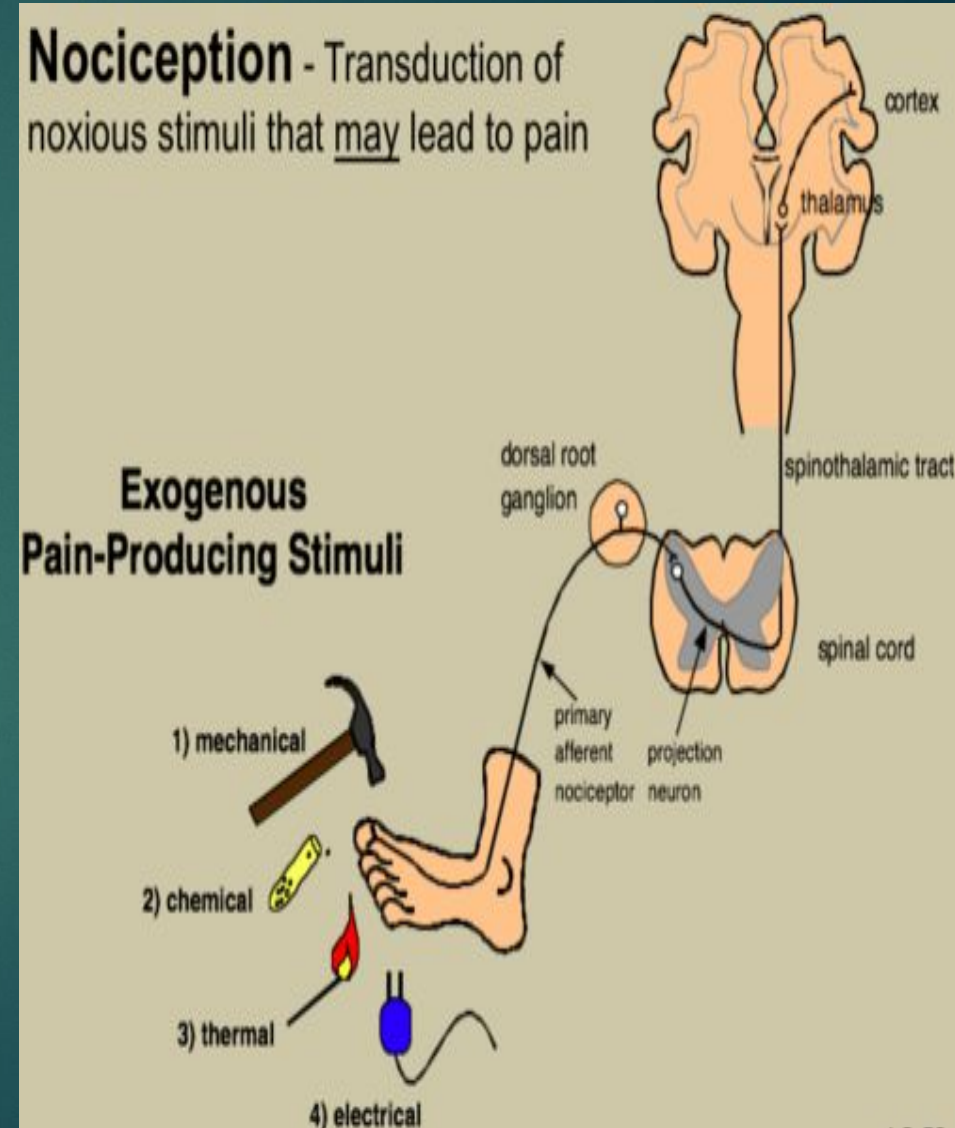
Receptors for pain sensation are the **free nerve endings** which are distributed throughout the body.

STIMULATION OF PAIN RECEPTORS

Pain can be elicited by various stimuli. The stimuli that stimulate pain receptors are classified as:

- ❑ Mechanical pain stimuli
- ❑ Thermal pain stimuli
- ❑ Chemical pain stimuli

Fast pain is elicited by the mechanical & thermal stimuli, whereas slow pain can be elicited by all three types of stimuli.



CHEMICAL PAIN STIMULATION

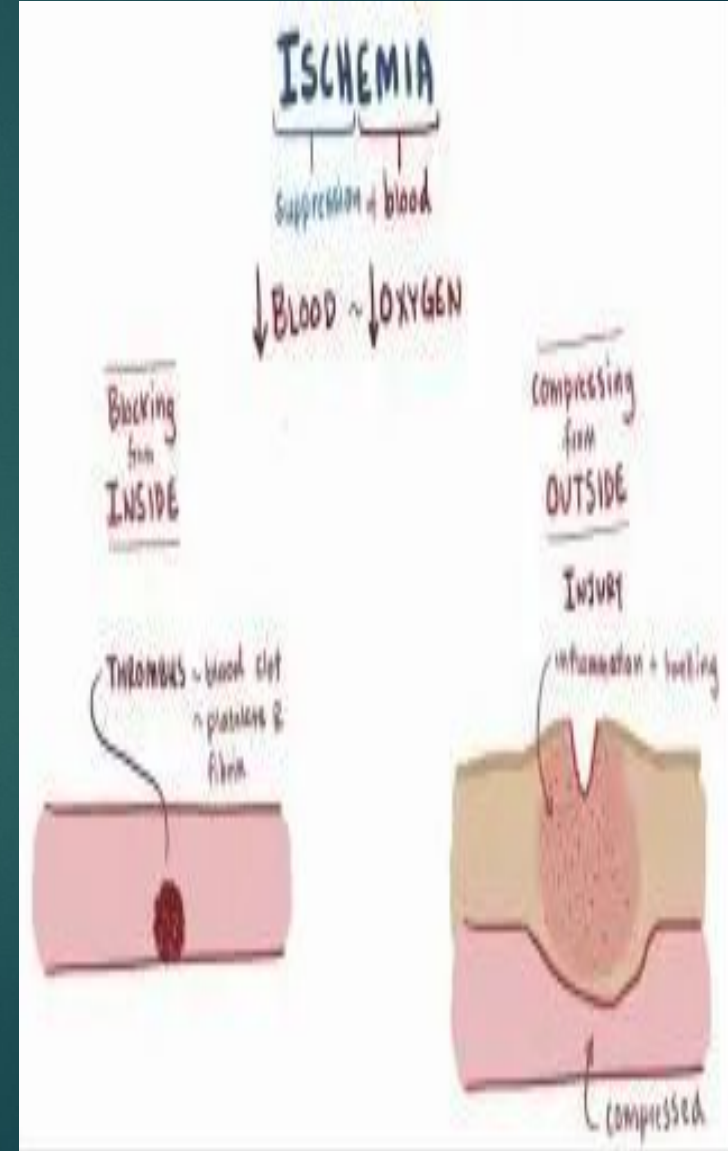
- Some of the chemicals that excite the chemical type of pain are **bradykinin**, **serotonin**, **histamine**, **potassium ions**, **acids**, **acetylcholine** & **proteolytic enzymes**.
- **Prostaglandins** & **substance P** can enhance the sensitivity of pain endings but do not directly stimulate them.

THERMAL PAIN STIMULATION

- On average, a person begins to perceive pain when the skin is heated above 45°C. This is the temperature at which the tissues begin to be damaged by heat.
- If the temperature remains above this level indefinitely, the tissues are eventually destroyed.

TISSUE ISCHEMIA PRODUCES PAIN

- When the blood flow to a tissue is blocked, the tissue often becomes very painful within a few minutes.
- The greater the rate of metabolism of the tissue, the more rapidly the pain appears.
- For example, if a blood pressure cuff is placed around the upper arm & inflated until the arterial blood flow ceases, exercise of forearm muscles can cause muscle pain within 15 to 20 seconds.
- In the absence of muscle exercise, pain may not be felt for 3 to 4 minutes even though the muscle blood flow remains zero.
- One of the suggested causes of pain during ischemia is the accumulation of large amounts of lactic acid in the tissues formed as a consequence of anaerobic metabolism.



MUSCLE SPASM PRODUCES PAIN

- ❑ Muscle spasm is also a common cause of pain & is the basis of many clinical pain syndromes.
- ❑ This pain results partially from the direct effect of muscle spasm on the stimulation of mechanosensitive pain receptors but also from the indirect effect of muscle spasm to compress the blood vessels & cause ischemia.
- ❑ Spasm also increases the rate of metabolism in the muscle tissue making the relative ischemia even greater & creating ideal conditions for the release of chemical pain inducing substances.



PATHWAYS FOR TRANSMISSION OF PAIN SIGNALS INTO THE CENTRAL NERVOUS SYSTEM

- Even though all pain receptors are free nerve endings, these ending use two separate pathways for the transmission of pain signals into the central nervous system.
 1. A fast-sharp pain pathway
 2. A slow-chronic pain pathway.

FAST(SHARP PAIN) FIBRES

- The fast-sharp pain signals are elicited by either mechanical or thermal pain stimuli.
- They are transmitted in the peripheral nerves to the spinal cord by **A-delta fibres** at velocities between **6 & 30 m/sec**.
- The sharp pain plays an important role in making the person react immediately to remove himself from the stimulus.

SLOW(CHRONIC PAIN) FIBRES

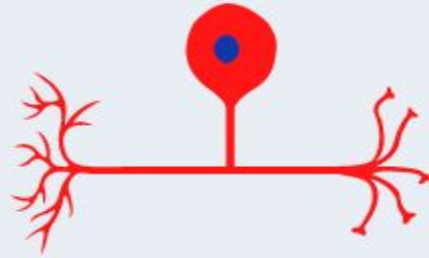
- The slow-chronic pain is elicited mostly by chemical types of pain stimuli but sometimes by persisting mechanical or thermal stimuli.
- This slow-chronic pain is transmitted to the spinal cord by **type-C fibres** at velocities between **0.5 & 2 m/sec**.
- The slow pain tends to become greater over time, eventually producing intolerable pain & making the person trying to relieve the cause of cause.

DOUBLE SYSTEM OF PAIN INNERVATION

Due to this double system of pain innervation, a sudden painful stimulus often gives a **double pain sensation**: a **fast-sharp pain** transmitted to the brain by **A-delta fibre pathway**, followed a second or so later by a **slow pain** that is transmitted via **C-fibre pathway**.

NOCICEPTORS

C-Fibers



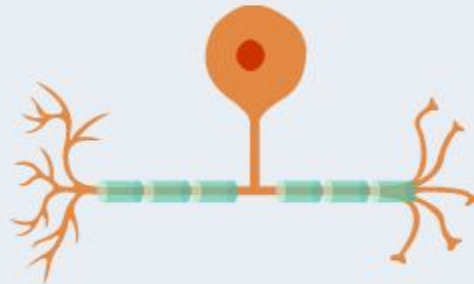
- Unmyelinated
- Slow Pain
- Low pH
- Heat
- Capsaicin-sensitive

Type I A δ -Fibers



- Myelinated
- Gross Touch
- Fast Pain
- High Heat (>53C)
- Capsaicin-insensitive

Type II A δ -Fibers



- Myelinated
- Gross Touch
- Fast Pain
- Low Heat (>43C)
- Capsaicin-sensitive

PAIN PATHWAYS IN THE SPINAL CORD & BRAIN STEM

On entering the spinal cord from the dorsal spinal roots, the pain fibres terminate on relay neurons in the dorsal horns. Here again, are the two pathways to transmit the pain signals to the brain:

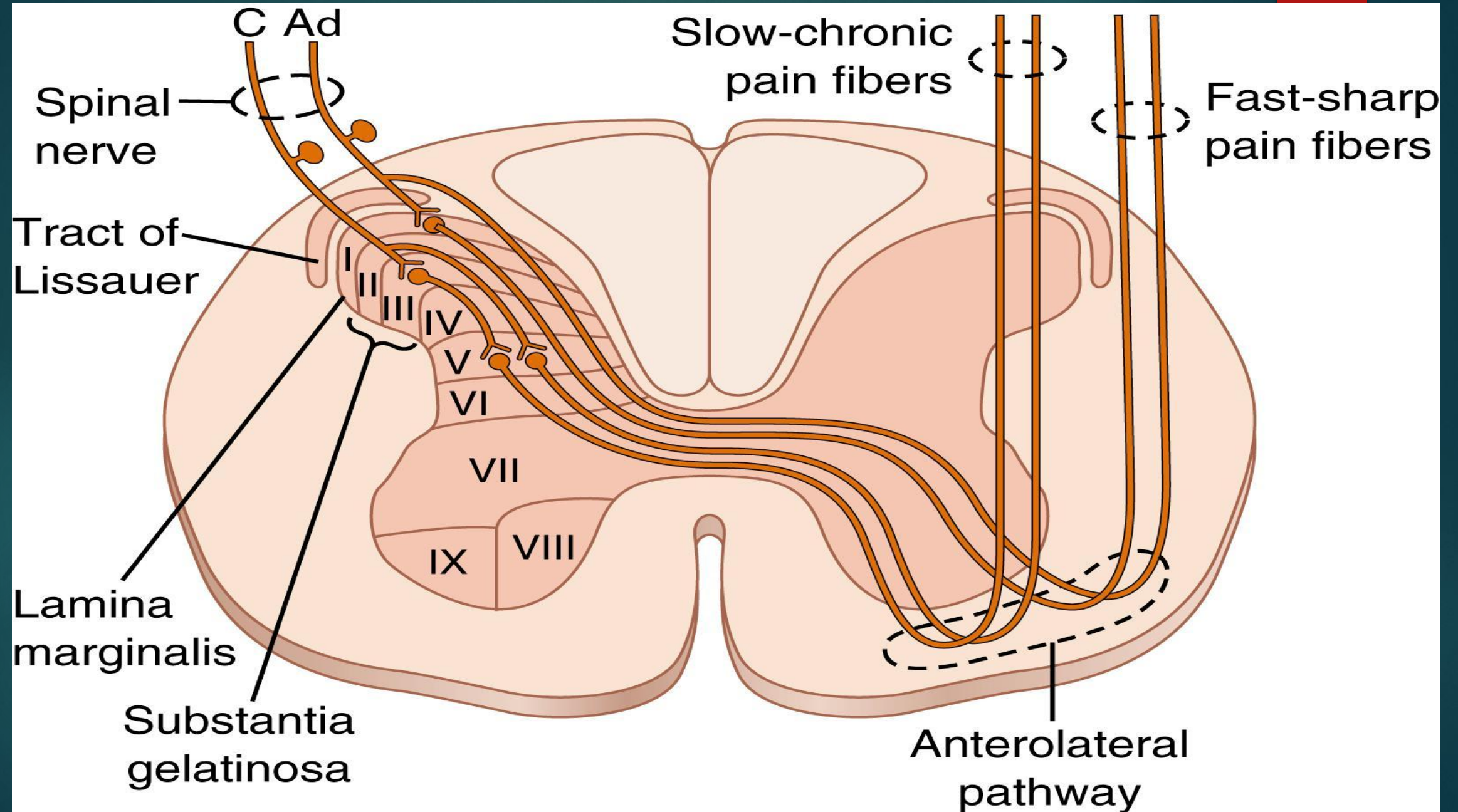
- ▣ **Neospinothalamic tract**(for fast-sharp pain)
- ▣ **Paleospinothalamic tract**(for slow-chronic pain)

1.NEOSPINOTHALAMIC PATHWAY FOR FAST PAIN

- ▣ The fast type(A-delta)pain fibers transmit mainly mechanical & acute thermal pain & terminate in lamina-I(lamina marginalis) of the dorsal horns to excite second-order neurons of neospinothalamic tract.

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- These second order neurons give rise to long fibers that cross immediately to the opposite side of the cord through the anterior commissure & then turn upward, passing to the brain in the anterolateral columns.
- A few fibers of the neospinothalamic tract terminate in the reticular areas of the brain stem while most pass all the way to the thalamus without interruption, terminating in the ventrobasal complex along with the dorsal column (medial lemniscal tract for tactile sensations).
- A few fibers also terminate in the posterior nuclear group of the thalamus.
- From these thalamic areas, the signals are transmitted to other basal areas of brain & also to the somatosensory cortex.

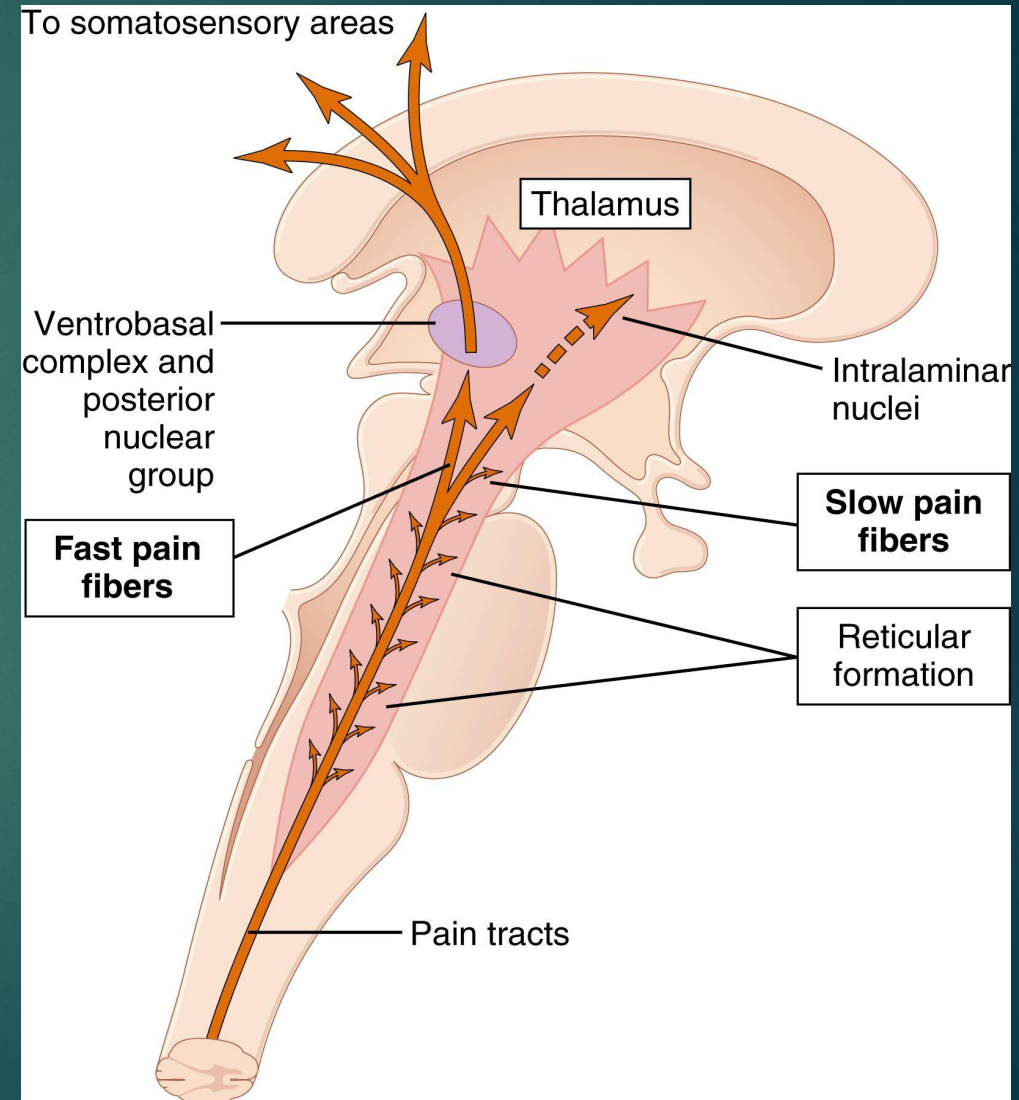


2. PALEOSPINOHALAMIC PATHWAY FOR SLOW PAIN

- The paleospinothalamic pathway transmits pain signals mainly from the peripheral slow(chronic) type-C fibers but also transmits some signals from type A-delta fibers.
- In this pathway, the peripheral fibers terminate in the spinal cord almost entirely in laminae II & III of the dorsal horns, collectively called substantia gelatinosa.
- Most of the signals then pass through one or more additional short fiber neurons within the dorsal horn before entering lamina V.
- Here the neurons give rise to long axons that mostly join the fibers from the fast pain pathway, passing first to the opposite side of cord through the anterior commissure & then upward to the brain in the anterolateral pathway.

CONT..

- The paleospinothalamic pathway terminates widely in the brain stem. Only 10% to 25% of the fibers pass all the way to the thalamus.
- Most fibers of this pathway terminates in one of these three lower regions of brain:(1) the reticular nuclei of medulla (2) the tectal area of mesencephalon or (3) the periaqueductal gray region surrounding the aqueduct of sylvius.
- From here the third order neurons transmit the pain signals upward into the intralaminar & ventrolateral nuclei of the thalamus & into certain portions of hypothalamus & other basal region of the brain.



NEUROTRANSMITTERS INVOLVED IN PAIN TRANSMISSION

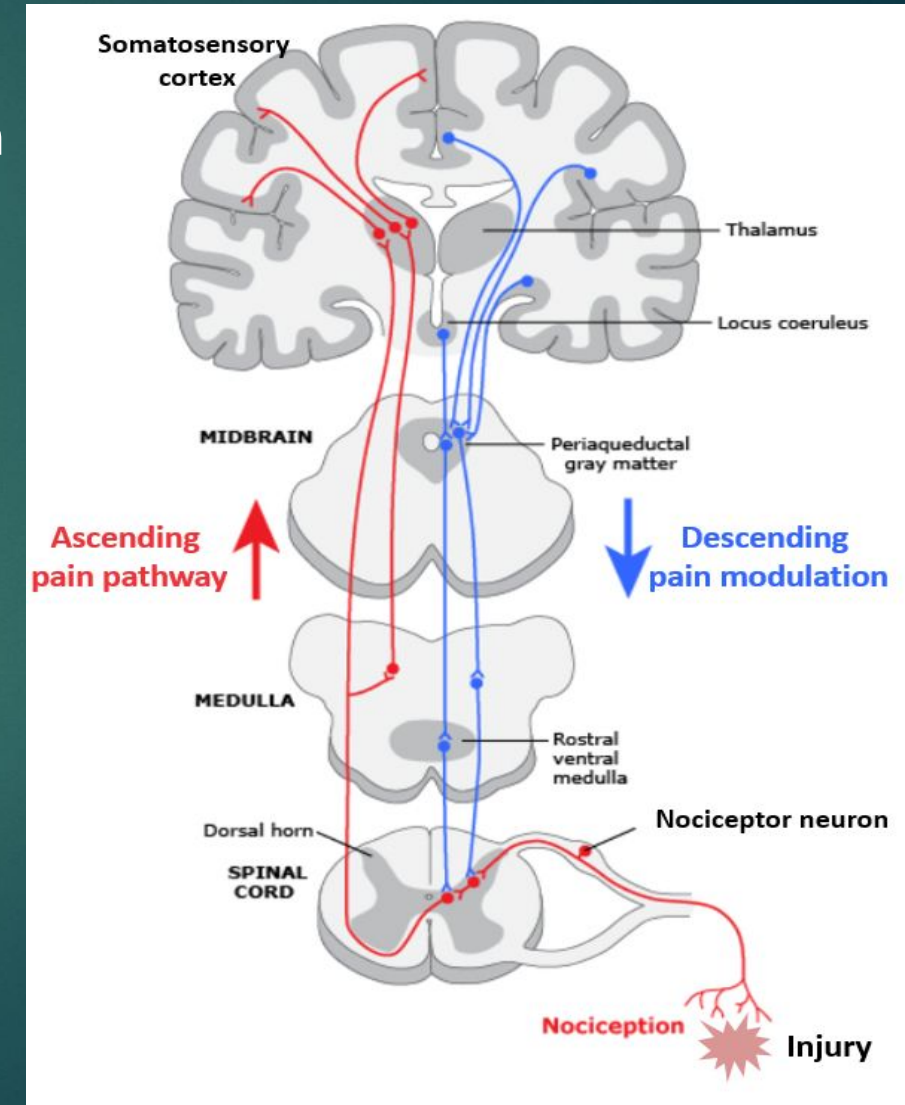
- Glutamate & substance P are the neurotransmitters secreted by pain nerve endings.
- **A-delta afferent fibers** which transmit impulses of fast pain secrete **glutamate**.
- The **type-C fibers** which transmit impulses of slow pain secrete **substance-P**.

ANALGESIA SYSTEM

- Analgesia system also called **endogenous analgesia system** is the pain control system of the body.
- This system present in brain provides short-term relief from pain.
- Analgesia system blocks the synaptic transmission of pain sensation in spinal cord & causes relief from pain.
- Analgesic drugs act through this system & provide a controlled relief from pain.

ANALGESIC PATHWAY

- Analgesic pathway is considered as **descending pain pathway**.
- Afferent fibers that transmit the pain sensation to the brain constitutes the **ascending pain pathway**.



ANALGESIC PATHWAY INHIBITS PAIN TRANSMISSION

- Fibers of analgesic pathway arise from frontal lobe of cerebral cortex & hypothalamus.
- These fibers terminate in the gray matter surrounding the third ventricle & aqueduct of sylvius.
- Fibers from this point descend down to brain stem & terminate on:
 - I. ***Nucleus raphe magnus***, situated in reticular formation of lower pons & upper medulla.
 - II. ***Nucleus reticularis***, paragigantocellularis situated in medulla.
- Fibers from these reticular nuclei descend through lateral white column of spinal cord & reach the synapses in the afferent pain pathway in anterior gray horn(A-delta fibers synapse with the neurons of marginal nucleus & type C fibers synapse with the neurons of substantia gelatinosa.
- At this synaptic level, analgesic fibers release neurotransmitters & inhibit the pain transmission to the brain.

GATE CONTROL THEORY

Gate control theory was proposed by Ronald Melzack & Patrick Wall in 1965 to explain pain suppression.

According to this theory afferent pain fibers are blocked by gate mechanism. If the gate is opened pain is felt & if the gate is closed pain is suppressed.

MECHANISM OF GATE CONTROL AT SPINAL LEVEL

- When pain stimulus is applied to any part of body, the receptors of other sensations like touch are also stimulated.
- When all these impulses reach the spinal cord, the fibers of touch sensation send collaterals to the neurons of pain pathway(i.e. neurons of marginal nucleus & substantia gelatinosa).
- Impulses of touch sensation passing through these collaterals inhibit the release of glutamate & substance P from pain fibers.
- This closes the gate & transmission of pain sensation is blocked.

OROFACIAL FACIAL PAIN

Orofacial pain refers to the pain perceived in the face, mouth, jaws or related structures.

- Orofacial pain may arise from dental, musculoskeletal, neurologic or systemic causes.
- It is complex due to overlapping innervation & referred pain patterns.

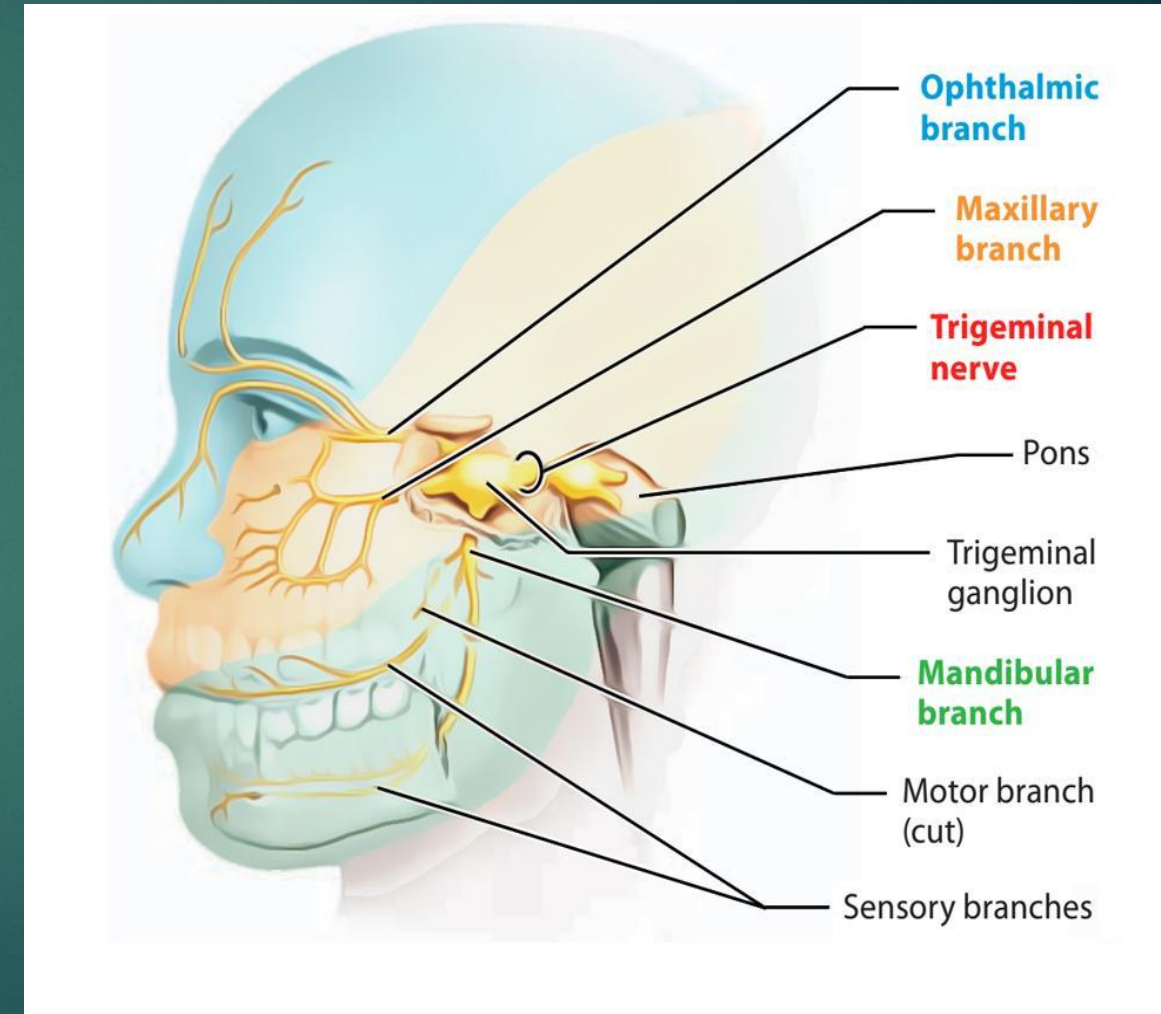
ANATOMY OF OROFACIAL PAIN

- **Trigeminal nerve** (CN-V) is the main sensory nerve of the face.
- Branches of trigeminal nerve are **ophthalmic** (V_1), **maxillary** (V_2) & **mandibular** (V_3).



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- ❑ Ophthalmic branch of trigeminal nerve innervates the forehead & eyes.
- ❑ Sensory innervation of middle face & upper teeth comes from the maxillary branch of trigeminal nerve.
- ❑ Mandibular branch gives sensory innervation to lower jaw & lower teeth & motor supply to the muscles involved in biting & chewing.



CLASSIFICATION OF OROFACIAL PAIN

ODONTOGENIC PAIN

- Most common type of orofacial pain usually seen in dental clinics.
- Common causes include **dental caries**, **pulpitis** or **periapical abscess** or trauma.
- Pain is usually well localized, sharp & throbbing in nature triggered by thermal stimuli(e.g. tea, icecream, cold water).

MUSCULOSKELETAL PAIN

- Associated with muscles & joints.
- Common condition is the **temporomandibular joint disorder**.
- Characteristic features include dull aching pain associated with jaw stiffness, clicking sound & limited mouth opening.

CONT..

NEUROPATHIC PAIN

- Caused by nerve injury or dysfunction.
- ▣ **Trigeminal neuralgia** is the common example of neuropathic pain.
- Characteristic features include sudden, electric shock-like pain triggered by light touch or movement.

NEUROVASCULAR PAIN

- Type of orofacial pain associated with vascular changes.
- Most common example of neurovascular pain is **migraine**.
- Characteristic features include pulsating pain associated with nausea & photophobia(sensitivity to light).

CONT..

PSYCHOGENIC PAIN

- Type of orofacial pain which has no specific physical cause.
- Usually linked with emotional or psychological factors.
- Patients may have anxiety or depression.
- Management involves psychological support.

REFERRED PAIN

- Referred pain can be defined as the pain felt in a different location from its site of origin.
- Examples include heart pain felt in the jaw or pain due to sinus infection felt in the upper tooth.

POST ASSESSMENT

Q1. Which type of pain is typically continuous & poorly localized:

- a) Neuropathic pain
- b) Somatic pain
- c) Visceral pain
- d) Referred pain

Q2. The main ascending pathway for pain transmission is:

- a) Corticospinal tract
- b) Vestibulospinal tract
- c) Spinothalamic tract
- d) Dorsal column

CONT..

Q3. Sharp, shooting pain triggered by light touch is characteristic of:

- a) Migraine
- b) Sinusitis
- c) Dental abscess
- d) Trigeminal neuralgia

Q4. Pain worsened by bending forward is most suggestive of:

- a) Dental caries
- b) Sinusitis
- c) Neuralgia
- d) TMJ disorder